YAMEN AL-KABBANI

Robotics & Control Engineer | Autonomous Systems · ROS2 · Mechatronics

Syria · +963 954 294 126 · Kabaneyamen@gmail.com · Open to relocation across the Middle East

LinkedIn: [add your URL] · GitHub: [add your URL] · Portfolio / Demo videos: [add your link]

PROFESSIONAL SUMMARY

Robotics & Control Engineer (B.Sc., 2026) specializing in autonomous robotic systems, ROS2, and mechatronic design. Designed and built a **fully autonomous 6-DOF vision-guided robotic arm** — depth-camera perception with ROS2 containerized in Docker on a Raspberry Pi — awarded a **perfect 100% graduation grade**. National robotics champion (**1st place — ARC 2026 Sumo, Syria**) and hands-on trainer of 30+ students and multiple competition teams. Currently developing a field-tested solar-panel-cleaning robot for utility-scale solar farms. Available immediately for R&D robotics roles.

TECHNICAL SKILLS

- **Robotics & Autonomy:** ROS2, autonomous manipulation, robotic-arm design, systems integration, sensor/actuator integration
- **Programming:** Python, C++, Arduino
- **Perception & AI:** Computer Vision (OpenCV), depth-camera 3D perception, AI-assisted development
- **Embedded & Hardware:** Raspberry Pi, multiple microcontroller families, serial-bus servos, motor & motion control
- **Mechanical & Prototyping:** SolidWorks (3D mechanical design), 3D printing, design-for-manufacture
- **Tools & DevOps:** Docker, Linux, Git

KEY PROJECTS

Autonomous 6-DOF Robotic Arm — Graduation Project Grade: 100%

- Designed and built a **fully autonomous 6-degree-of-freedom robotic arm** with an end-to-end perception-to-actuation pipeline.
- Integrated a depth camera for 3D object perception; ran the full software stack on **ROS2, containerized with Docker on a Raspberry Pi**.
- Implemented serial-bus servo control for precise, coordinated multi-joint motion — achieving full autonomy from sensing through planning to actuation.

Solar-Panel Cleaning Robot — Near-MVP, Field-Tested (Entrepreneurial) 2025 – Present

- Developing a remote-controlled robot to clean photovoltaic panels across **utility-scale solar farms (3,000+ panels)**.
- Validated in real operating conditions; finalizing a market-ready MVP with a clear path to commercialization in the solar-O&M sector.

Competition Robots — Sumo & Robotic Arms

- Designed and programmed multiple autonomous Sumo robots — **1st place in Syria (ARC 2026)**; built additional robotic arms for training and demonstration.

EXPERIENCE

Robotics Trainer — National Center for Distinguished Students 2024

- Selected to train Syria's top-performing students in robotics design, programming, and competition strategy.
- Coached **4+ competitive robotics teams**, several advancing to national-level events.

Robotics Trainer — Sera Robot Center 2023 – 2024

- Delivered hands-on robotics training to **30+ students** across multiple age groups, building skills in programming and mechatronics.
- Prepared teams for national and international competitions (WRO, ARC), mentoring them to top rankings.

COMPETITIONS & ACHIEVEMENTS

- **1st Place, Syria** — ARC 2026 Sumo Robot Competition
- **2nd Place, Syria** — RFG Robotics Competition
- **1st Place, Audience Award** — ARC Competition
- Coached student teams to top national rankings in WRO (World Robot Olympiad).

CERTIFICATIONS

- **WRO (World Robot Olympiad)** — Trainer & Judge Certifications
- **ARC Competition** — Trainer & Judge Certifications

EDUCATION

B.Sc. Informatics Engineering — Control & Robotics Engineering 2026

Syrian Private University · Graduation Project: 100%

LANGUAGES

Arabic (Native) · English (Fluent) — strong technical communication in both